

[DC] Improving Presence of Virtual Humans through Paralinguistics

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ABSTRACT

Social presence and plausibility are two of the most important constructs being studied in IEEE VR to improve user experience in virtual environments. However, virtual humans that are used in these virtual environments lack the capability of adaptive speech that directly influences social presence and plausibility of the virtual humans that are used since humans use adaptive speech during conversations. Virtual humans lack the adaptability of their speech in the manner that humans do. Research has been explored in how pitch, rate of speech, and intensity can be manipulated in human discourse. However, little research has been done in using adaptable speech to affect dialog with virtual humans. This dissertation attempts to create a framework and a system that enables adaptive virtual human speech. This includes using machine learning and user modeling to manipulate paralinguistic features such as pitch, rate of speech, pause duration, and intensity using Speech Synthesis Markup Language (SSML). Understanding the interplay of paralinguistic features between virtual human and human will improve the social presence of virtual humans. By achieving this, virtual interactions can be moved toward the dynamic level of human-to-human interactions, thus increasing the co-presence and plausibility of these virtual human characters.

Keywords: Virtual Reality, presence, paralinguistics, virtual humans

Index Terms: Human-centered computing—Interaction design; Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality

1 INTRODUCTION

We have the capability of placing a person in the middle of a simulated classroom, doctor's office, or alien planet using virtual reality. We can put a virtual human in front of them that looks so realistic they might swear it was another human present with them. Disbelief is suspended and plausibility is high for many people placed in front of many modern virtual humans. That is until the moment the virtual human opens their mouth and speaks using a computer-generated voice. Social presence begins to wane, and plausibility reduces as the person starts to realize (remember) it is a computer.

Academia and the computer industry continue to invest countless amounts of money and hours in creating more realistic looking graphics for virtual reality, whether it is on a desktop screen, head-mounted display, or cave system. Graphic processors are so powerful that they are now the "go to" for artificial intelligence. Yet, when we think about the speeches that have changed our lives or touched our hearts it is not what the person was wearing, the color of their eyes, the way they parted

their hair that we remember. We likely couldn't even see those things in detail. It was the words that were spoken and how those words were spoken that we remember. How those words were/are spoken is the study of paralinguistics – the qualities of a voice such as pitch, rate of speech, duration of pauses, and volume used when a human or virtual human is speaking. In the case of a virtual human using a computer-generated voice those voices are said to sound robotic. That robotic sound is not because better quality voices are required, but instead because the paralinguistic features of the voice do not change during a vocal utterance. It is my position that by improving the paralinguistic features used by virtual humans the social presence and plausibility of the virtual human can be increased. For my thesis I will be focusing on four questions: (1) What paralinguistic features studied in human voices can be used by virtual humans? (2) What roll does the voice used (e.g., Google en-US-Standard-A, Amazon Polly Joey, IBM Watson Michael) play in the perception of altering paralinguistic features? (3) Can paralinguistic features be used to alter the perceived personality traits of a virtual human? (4) Can paralinguistic features be used to alter the perceptions people have of a virtual human, such as education level, ethnicity, or social status? I will attempt to answer these questions as part of my research and dissertation.

2 LITERATURE REVIEW

The vocal characteristics of human speech has been a much-studied aspect of psychology and provides a foundation for how people will relate to similar characteristics of computer-generated speech. Research on the perceived personal attributions of a disembodied speaker found that higher-pitched voices were perceived as less truthful, more nervous, and less potent physically (smaller frame and stature) while a lower rate of speech was perceived as less truthful, less persuasive, and increased passivity [1]. Research on the effects rate of speech, volume and pitch had on dominance judgements found that volume was positively correlated to dominance, while rate of speech was negatively correlated to dominance [2]. Pitch for the voice was found to positively correlate to dominance for male speakers but had no significance for female speakers [2]. A study in 2012 looked at the effects of pitch on perceived leadership ability and found that a lower pitched voice increased the perception of leadership in both male and female voices [3]. Given this prior work in the study of voice, I hope to discern the vocal attributes that can be used in virtual human voices to produce human-like personality traits. I also hope to discover if the paralinguistics for different human genders apply to engendered virtual humans.

Researchers have been developing methods of tailoring virtual humans to users based on their personality and the subject domain. Research on whether computer-generated speech could have perceivable personality attributes found that extroverted users preferred computer-generated speech that exhibited vocal qualities attributed to extroverts, such as higher rate of speech and more varied pitch [4]. Additionally, this research found that introverted users held no preference toward either extrovert or introvert speech qualities [4]. A study examining the effects computer-generated voices and 3D avatars have on a user's perception of a computer system found that voiced agents are

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perceived as more playful and embodied agents increase the user's sense of telepresence [5]. Research on the effects gender matching has on user to computer-generated voices found that female users aligned more to male voices, while male users were equally aligned with either voice gender [6]. Given this prior work in computer-generated speech, I hope to increase understanding in how perceptions of virtual human personality can be tailored through their voice.

3 PROPOSED RESEARCH

I am proposing completing my research across three phases. During the first phase of my research, I plan to recreate existing human paralinguistic studies using voice actors. While existing studies from the fields of psychology and human behavior identify many of the paralinguistic features associated with emotions, personality traits, and behaviors, these studies do not provide analysis at a level of depth I feel is necessary to identify the relationship between each paralinguistic feature. Additionally, different genders, races, and cultures [7] have different perceptions of these features and this demographic information should be identified. The second phase of my research will use the analysis from existing research and studies run during the first phase to attempt to recreate the same perceptions and behaviors using virtual humans. These studies will attempt to identify physical traits of virtual humans that best correspond with different vocal qualities by using the previously created recordings from voice actors. Perceptions of the virtual human and voice will be catalogued. The third phase of my research will attempt to recreate the perceptions of each virtual human seen in the second phase using computer-generated voices instead of the recordings from voice actors. The paralinguistic features of each voice will be catalogued and then altered to match those of the second phase. It is my hope that through this process a deeper understanding of the interplay between the different paralinguistic features, the physical characteristics of the virtual human, and the listener can be identified.

4 CONCLUSION

For the first phase, it seems like it would be easy to get lost in continuing to research more and more variations of features or previous studies. What level is an appropriate goal? How do I identify a point to move to the virtual human phase?

For subsequent phases, how focused or broad should matching with the first given the variety of physical characteristics that can be changed for a virtual human and number of voices available? Should I focus instead on ways to procedurally (programmatically) match characteristics to a voice? Or should this be done one personality trait at a time?

It is my intention to contribute to the understanding voice and vocal characteristics have on the social presence of a virtual human. Through the embodiment of enhanced personality through voice and increased variation in speech patterns for a virtual human, I hope to advance the area of co-presence in VR. The increase in social presence and co-presence for virtual humans should contribute to plausibility for VR involving speaking virtual humans.

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